

CN PHASE 1 DOCUMENTATION

MAC 4

1. Backend B working locally

```
saumyamishra@Saumyas-MacBook-Pro-3 cn-backend-b % curl -i https://app.cn-capstone.test:8443/api/status
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 10:04:50 GMT
Content-Type: application/json
Content-Length: 32
Connection: keep-alive
Cache-Control: max-age=60
ETag: "B-v1"
X-Backend: B

{"backend": "B", "status": "ok"}
saumyamishra@Saumyas-MacBook-Pro-3 cn-backend-b %
```

Backend B Health Check: The `/api/status` endpoint returns HTTP 200 OK and identifies Backend B using the `X-Backend: B` header. The response also contains caching headers (`ETag` and `Cache-Control`).

2. Cache validation / 304 response

```
saumyamishra@Saumyas-MacBook-Pro-3 cn-backend-b % curl -i \
-H 'If-None-Match: "B-v1"' \
https://app.cn-capstone.test:8443/api/status
HTTP/1.1 304 Not Modified
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 10:24:18 GMT
Connection: keep-alive
ETag: "B-v1"
Cache-Control: max-age=60
X-Backend: B
```

HTTP Cache Validation: A subsequent request containing `If-None-Match: "B-v1"` receives `304 Not Modified`, demonstrating ETag-based cache validation and avoiding retransmission of unchanged content.

3. DNS resolution

dns.qry.name == "app.cn-capstone.test"							
No.	Time	Source	Destination	Protocol	Length	Info	
37...	25.3327...	Saumyas-MacB...	10.7.9.245	DNS	91	Standard query 0xb210 A app.cn-capstone.test OPT	
37...	25.3784...	10.7.9.245	Saumyas-MacB...	DNS	107	Standard query response 0xb210 A app.cn-capstone.test A 10.7.5.53 OPT	

DNS Resolution: Wireshark captures the DNS query for `app.cn-capstone.test` and the corresponding response mapping the hostname to `10.7.5.53`, confirming successful name

resolution.

```
saumyamishra@Saumyas-MacBook-Pro-3 cn-backend-b % dig @10.7.9.245 app.cn-capstone.test

; <<>> DiG 9.10.6 <<>> @10.7.9.245 app.cn-capstone.test
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 8533
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;app.cn-capstone.test.      IN      A

;; ANSWER SECTION:
app.cn-capstone.test.      30      IN      A      10.7.5.53

;; Query time: 6 msec
;; SERVER: 10.7.9.245#53(10.7.9.245)
;; WHEN: Thu Oct 01 15:34:28 IST 2026
;; MSG SIZE rcvd: 65
```

DNS Verification using dig: The DNS server 10.7.9.245 successfully resolves app.cn-capstone.test to 10.7.5.53.

4. TCP 3-way handshake

The image displays a Wireshark packet capture of a TCP 3-way handshake. The top pane shows a list of packets, with the first three packets (331, 332, and 333) highlighted in red, indicating they are part of the handshake. The second pane shows the details of the selected packet (331), which is a SYN packet from the client to the server. The third pane shows the packet bytes in hexadecimal and ASCII.

Packet 331: SYN

- Time: 0.000000000
- Source: 10.7.9.245
- Destination: 10.7.5.53
- Protocol: TCP
- Length: 60
- Info: Seq=331, Win=0, Len=0, TSval=3308927948, TSecr=66467553

Packet 332: SYN-ACK

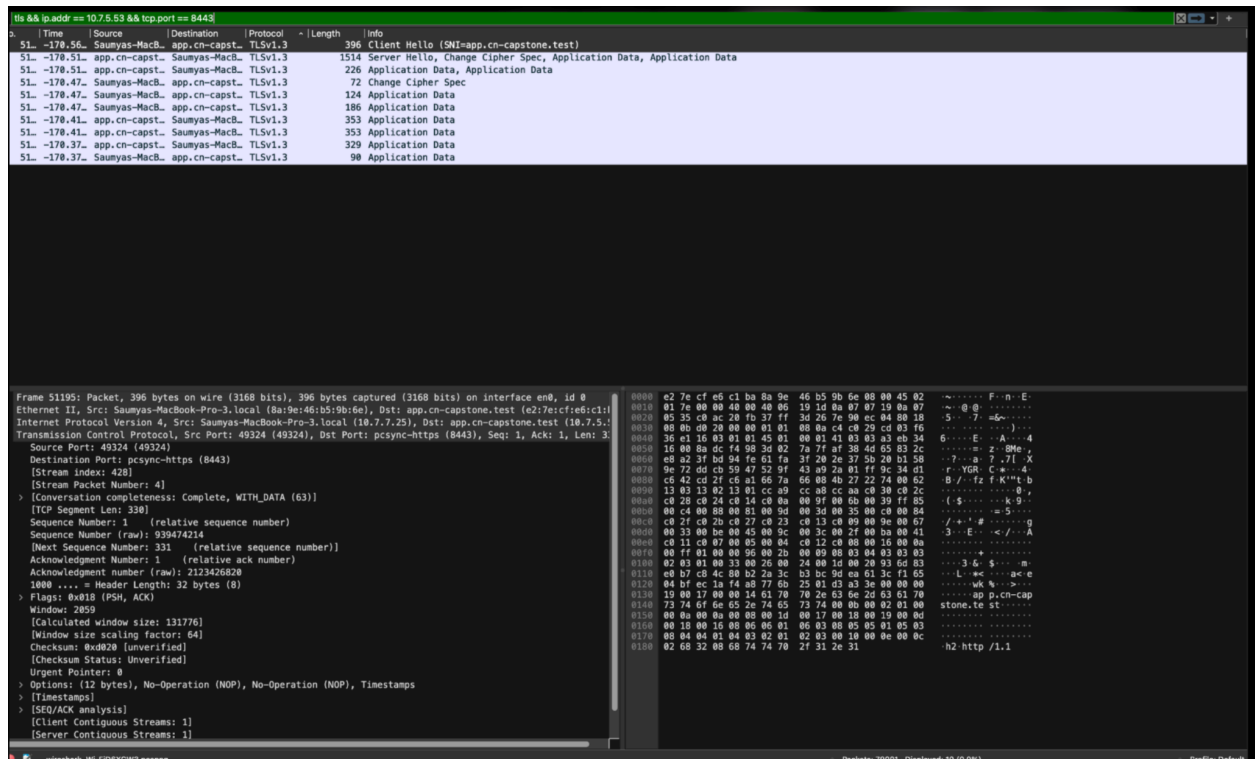
- Time: 0.000000000
- Source: 10.7.5.53
- Destination: 10.7.9.245
- Protocol: TCP
- Length: 60
- Info: Seq=1609, Win=0, Len=0, TSval=66467561, TSecr=3308927948

Packet 333: ACK

- Time: 0.000000000
- Source: 10.7.9.245
- Destination: 10.7.5.53
- Protocol: TCP
- Length: 60
- Info: Seq=331, Win=0, Len=0, TSval=3308927948, TSecr=66467561

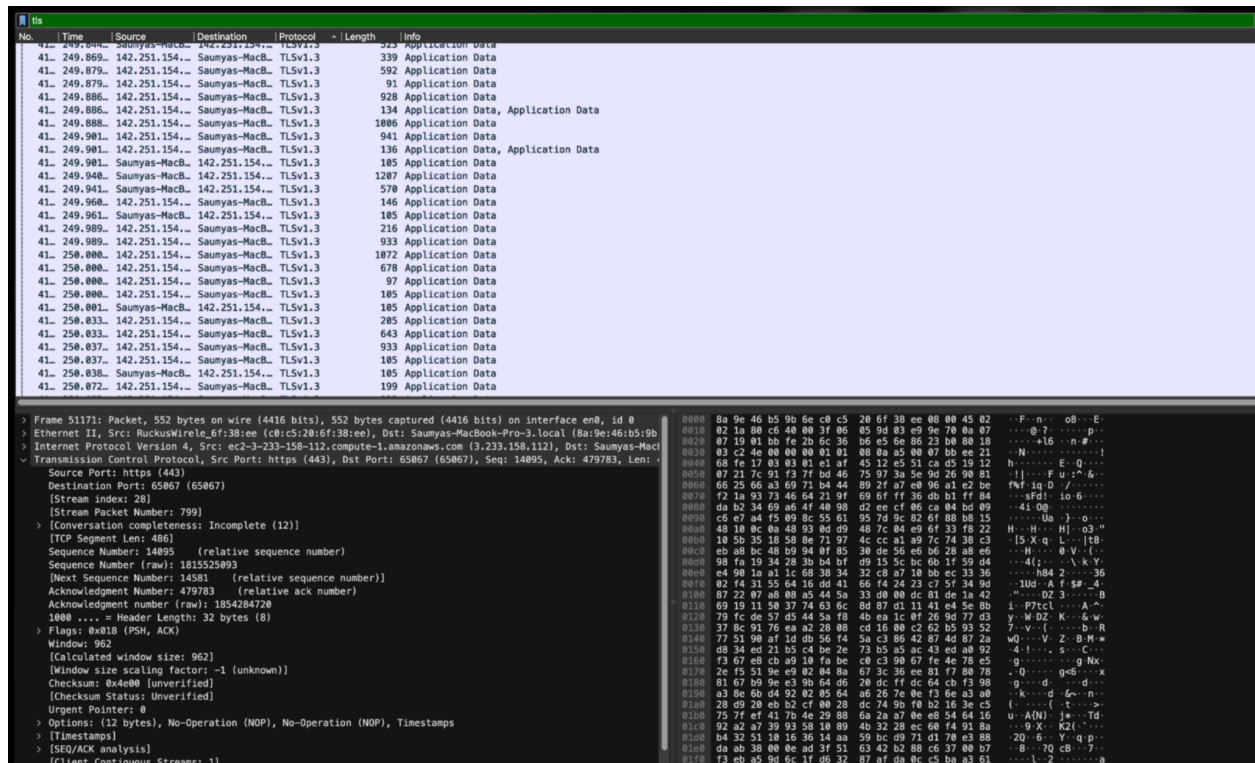
TCP Three-Way Handshake: Wireshark shows the SYN, SYN-ACK and ACK sequence establishing a reliable TCP connection between the client and the HTTPS server on port 8443.

5. TLS/HTTPS handshake



TLS 1.3 Handshake: Wireshark captures the TLS 1.3 connection establishment, including the Client Hello and Server Hello messages, followed by encrypted Application Data. This demonstrates the establishment of secure HTTPS communication.

6. Encrypted HTTPS Application Data



Encrypted HTTPS Traffic: Wireshark captures TLS 1.3 Application Data packets exchanged between the client and server. The application payload is encrypted and cannot be directly read, demonstrating secure data transmission.

7.Load Balancing

```
saumyamishra@Saumyas-MacBook-Pro-3 cn-backend-b % for i in {1..20}; do
  curl -s -D - \
    --resolve app.cn-capstone.test:8443:10.7.5.53 \
    https://app.cn-capstone.test:8443/api/status \
    -o /dev/null | grep X-Backend
done
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
```

Load Balancing Verification: Repeated requests to the application endpoint alternate between Backend A and Backend B, as shown by the **X-Backend** response header. This demonstrates successful distribution of requests across both backend servers.